

UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

- ☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the fiscal year ended December 30, 2023.
or
☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the transition period from _____ to _____.

Commission File Number: 000-06217



INTEL CORPORATION

(Exact name of registrant as specified in its charter)

Delaware

(State or other jurisdiction of incorporation or organization)

2200 Mission College Boulevard, Santa Clara, California

(Address of principal executive offices)

94-1672743

(I.R.S. Employer Identification No.)

95054-1549

(Zip Code)

Registrant's telephone number, including area code: **(408) 765-8080**

Securities registered pursuant to Section 12(b) of the Act:

Title of each class
Common stock, \$0.001 par value

Trading symbol
INTC

Name of each exchange on which registered
Nasdaq Global Select Market

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every interactive data file required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer", "accelerated filer", "smaller reporting company", and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large Accelerated Filer
☒

Accelerated Filer
☐

Non-Accelerated Filer
☐

Smaller Reporting Company
☐

Emerging Growth Company
☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

The aggregate market value of voting and non-voting common equity held by non-affiliates of the registrant as of June 30, 2023, based upon the closing price of the common stock as reported by the Nasdaq Global Select Market on such date, was \$140.0 billion. 4,228 million shares of common stock were outstanding as of January 19, 2024.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant's proxy statement related to its 2024 Annual Stockholders' Meeting to be filed subsequently are incorporated by reference into Part III of this Form 10-K. Except as expressly incorporated by reference, the registrant's proxy statement shall not be deemed to be part of this report.

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Organization of Our Form 10-K

The order and presentation of content in our Form 10-K differs from the traditional SEC Form 10-K format. Our format is designed to improve readability and better present how we organize and manage our business. See "Form 10-K Cross-Reference Index" within the Financial Statements and Supplemental Details for a cross-reference index to the traditional SEC Form 10-K format.

We have defined certain terms and abbreviations used throughout our Form 10-K in "Key Terms" within the Financial Statements and Supplemental Details.

The preparation of our Consolidated Financial Statements is in conformity with US GAAP. Our Form 10-K includes key metrics that we use to measure our business, some of which are non-GAAP measures. See "Non-GAAP Financial Measures" within MD&A for an explanation of these measures and why management uses them and believes they provide investors with useful supplemental information.

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Forward-Looking Statements

This Form 10-K contains forward-looking statements that involve a number of risks and uncertainties. Words such as "accelerate", "achieve", "aim", "ambitions", "anticipate", "believe", "committed", "continue", "could", "designed", "estimate", "expect", "forecast", "future", "goals", "grow", "guidance", "intend", "likely", "may", "might", "milestones", "next generation", "objective", "on track", "opportunity", "outlook", "pending", "plan", "position", "possible", "potential", "predict", "progress", "ramp", "roadmap", "seek", "should", "strive", "targets", "to be", "upcoming", "will", "would", and variations of such words and similar expressions are intended to identify such forward-looking statements, which may include statements regarding:

- our business plans and strategy and anticipated benefits therefrom, including with respect to our IDM 2.0 strategy, our Smart Capital strategy, our partnership with Brookfield, the transition to an internal foundry model, updates to our reporting structure, and our AI strategy;
- projections of our future financial performance, including future revenue, gross margins, capital expenditures, and cash flows;
- projected costs and yield trends;
- future cash requirements, the availability, uses, sufficiency, and cost of capital resources, and sources of funding, including for future capital and R&D investments and for returns to stockholders, such as stock repurchases and dividends, and credit ratings expectations;
- future products, services, and technologies, and the expected goals, timeline, ramps, progress, availability, production, regulation, and benefits of such products, services, and technologies, including future process nodes and packaging technology, product roadmaps, schedules, future product architectures, expectations regarding process performance, per-watt parity, and metrics, and expectations regarding product and process leadership;
- investment plans and impacts of investment plans, including in the US and abroad;
- internal and external manufacturing plans, including future internal manufacturing volumes, manufacturing expansion plans and the financing therefor, and external foundry usage;
- future production capacity and product supply;
- supply expectations, including regarding constraints, limitations, pricing, and industry shortages;
- plans and goals related to Intel's foundry business, including with respect to anticipated customers, future manufacturing capacity and service, technology, and IP offerings;
- expected timing and impact of acquisitions, divestitures, and other significant transactions, including the sale of our NAND memory business;
- expected completion and impacts of restructuring activities and cost-saving or efficiency initiatives
- future social and environmental performance goals, measures, strategies, and results;
- our anticipated growth, future market share, and trends in our businesses and operations;
- projected growth and trends in markets relevant to our businesses;
- anticipated trends and impacts related to industry component, substrate, and foundry capacity utilization, shortages, and constraints;
- expectations regarding government incentives;
- future technology trends and developments, such as AI;
- future macro environmental and economic conditions;
- geopolitical tensions and conflicts and their potential impact on our business;
- tax- and accounting-related expectations;
- expectations regarding our relationships with certain sanctioned parties; and
- other characterizations of future events or circumstances.

Such statements involve many risks and uncertainties that could cause our actual results to differ materially from those expressed or implied, including those associated with:

- the high level of competition and rapid technological change in our industry;
- the significant long-term and inherently risky investments we are making in R&D and manufacturing facilities that may not realize a favorable return;
- the complexities and uncertainties in developing and implementing new semiconductor products and manufacturing process technologies;
- our ability to time and scale our capital investments appropriately and successfully secure favorable alternative financing arrangements and government grants;
- implementing new business strategies and investing in new businesses and technologies;
- changes in demand for our products;
- macroeconomic conditions and geopolitical tensions and conflicts, including geopolitical and trade tensions between the US and China, the impacts of Russia's war on Ukraine, tensions and conflict affecting Israel, and rising tensions between the US and Taiwan;
- the evolving market for products with AI capabilities;

- our complex global supply chain, including from disruptions, delays, trade tensions and conflicts, or shortages;
- product defects, errata and other product issues, particularly as we develop next-generation products and implement next-generation manufacturing process technologies;
- potential security vulnerabilities in our products;
- increasing and evolving cybersecurity threats and privacy risks;
- IP risks including related litigation and regulatory proceedings;
- the need to attract, retain, and motivate key talent;
- strategic transactions and investments;
- sales-related risks, including customer concentration and the use of distributors and other third parties;
- our significantly reduced return of capital in recent years;
- our debt obligations and our ability to access sources of capital;
- complex and evolving laws and regulations across many jurisdictions;
- fluctuations in currency exchange rates;
- changes in our effective tax rate;
- catastrophic events;
- environmental, health, safety, and product regulations;
- our initiatives and new legal requirements with respect to corporate responsibility matters; and
- other risks and uncertainties described in this report and our other filings with the SEC.

Given these risks and uncertainties, readers are cautioned not to place undue reliance on such forward-looking statements. Readers are urged to carefully review and consider the various disclosures made in this Form 10-K and in other documents we file from time to time with the SEC that disclose risks and uncertainties that may affect our business.

Unless specifically indicated otherwise, the forward-looking statements in this Form 10-K do not reflect the potential impact of any divestitures, mergers, acquisitions, or other business combinations that have not been completed as of the date of this filing. In addition, the forward-looking statements in this Form 10-K are based on management's expectations as of the date of this filing, unless an earlier date is specified, including expectations based on third-party information and projections that management believes to be reputable. We do not undertake, and expressly disclaim any duty, to update such statements, whether as a result of new information, new developments, or otherwise, except to the extent that disclosure may be required by law.

Note Regarding Third-Party Information

This Form 10-K includes market data and certain other statistical information and estimates that are based on reports and other publications from industry analysts, market research firms, and other independent sources, as well as management's own good faith estimates and analyses. Intel believes these third-party reports to be reputable, but has not independently verified the underlying data sources, methodologies, or assumptions. The reports and other publications referenced are generally available to the public and were not commissioned by Intel. Information that is based on estimates, forecasts, projections, market research, or similar methodologies is inherently subject to uncertainties, and actual events or circumstances may differ materially from events and circumstances reflected in this information.

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Availability of Company Information

We use our Investor Relations website, www.intc.com, as a routine channel for distribution of important, and often material, information about us, including our quarterly and annual earnings results and presentations, press releases, announcements, information about upcoming webcasts, analyst presentations, and investor days, archives of these events, financial information, corporate governance practices, and corporate responsibility information. We also post our filings on this website the same day they are electronically filed with, or furnished to, the SEC, including our annual and quarterly reports on Forms 10-K and 10-Q and current reports on Form 8-K, our proxy statements, and any amendments to those reports. All such information is available free of charge. Our Investor Relations website allows interested persons to sign up to automatically receive e-mail alerts when we post financial information and issue press releases, and to receive information about upcoming events. We encourage interested persons to follow our Investor Relations website in addition to our filings with the SEC to timely receive information about the company.



We are an industry leader

and a catalyst for technology innovation and products that revolutionize the way we live. We are committed to harnessing the breadth and scale of our reach to have a positive effect on business, society, and the planet.

Our purpose is to create world-changing technology that improves the life of every person on the planet.



Introduction to Our Business

We are committed to creating world-changing technology that improves the life of every person on the planet—we are the technology the world builds on. We have the opportunity to push the boundaries of what's possible and to create solutions to the world's biggest challenges. As technology permeates every aspect of our lives, we see an insatiable demand for processing power.

That is why we are opening our manufacturing network to the world and creating the resilient supply chain industries need. It's why we are bringing the full breadth of our silicon and software to bear by bringing AI everywhere—from the client to the data center to the edge. It's why we continue our relentless pursuit of Moore's Law to unlock innovation and spark new ecosystems.

Much as oil has defined geopolitics for the past five decades, technology supply chains and where semiconductors are built will define the next five decades. With one of the most geographically balanced supply chains across North America, Europe and Asia, we lead the way in creating open, end-to-end value chains that the US and Europe

seek for resiliency and security—and that, for the first time, gives the ecosystem a true alternative to other foundries.

We are at a pivotal moment in AI technology. AI is an incredibly powerful technology with untold potential, but it's still relatively immature. We must ensure AI technology advances responsibly.

Intel defines the spirit of Moore's Law as relentless innovation and pursuit of exponential leaps in computing power—all in close collaboration with our customers. But we're not finished yet. Continuous innovation is the cornerstone of Moore's Law.

We do this because we are committed to being an excellent partner for the next era of compute: creating trusted environments, collaboratively innovating and delivering exceptional engineering, from silicon to services.

**The possibilities are limitless.
It starts with Intel.**

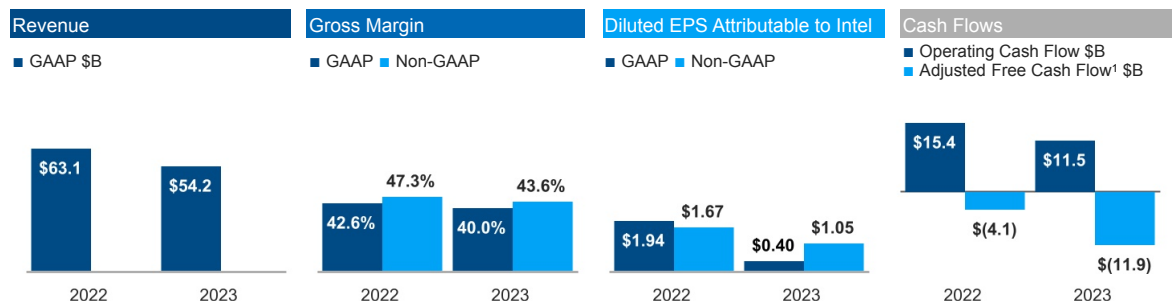


"Semiconductors are essential to maintaining and enabling modern society and there are infinite possibilities as we enter the age of AI. Our strategy for reclaiming process and product leadership, bringing AI everywhere and driving a resilient, diverse and balanced supply chain, puts Intel in the position to help define the future of technology."

— Pat Gelsinger, Intel CEO

A Year in Review

2023 revenue was \$54.2 billion, down \$8.8 billion, or 14%, from 2022. CCG revenue decreased 8% due to lower notebook and desktop volume from lower demand across market segments, partially offset by increased volume in the second half of the year as customer inventory levels normalized compared to higher levels in the first half. Notebook ASPs decreased due to a higher mix of small core products combined with a higher mix of older generation products. This was partially offset by higher desktop ASPs due to an increased mix of product sales to the commercial and gaming market segments. DCAI revenue decreased 20% due to lower server volume resulting from a softening CPU data center market, partially offset by higher ASPs from a lower mix of hyperscale customer-related revenue and a higher mix of high core count products. NEX revenue decreased 31% as customers tempered purchases to reduce inventories and adjust to a lower demand environment across product lines. We invested \$16.0 billion in R&D, made capital investments of \$25.8 billion, and had \$11.5 billion in cash from operations and \$(11.9) billion of adjusted free cash flow.



\$54.2B	40.0%	43.6%	\$0.40	\$1.05	\$11.5B	\$(11.9)B
GAAP	GAAP	non-GAAP ¹	GAAP	non-GAAP ¹	GAAP	non-GAAP ¹
Revenue down 14% from 2022 revenue	Gross margin down 2.6 pts from 2022	Gross margin down 3.7 pts from 2022	Diluted EPS down \$1.54 or 79% from 2022	Diluted EPS down \$0.62 or 37% from 2022	Operating cash flow down \$4.0B or 26% from 2022	Adjusted free cash flow down \$7.8B or 191% from 2022
Lower revenue in CCG, DCAI, and NEX.	Lower GAAP gross margin from lower revenue, higher unit cost, and higher excess capacity charges, partially offset by the sell-through of previously reserved inventory, lower inventory reserves taken in 2023, lower product ramp costs, and the absence of one-time charges recognized in 2022.		Lower GAAP EPS from lower gross margin and absence of one-time gains recognized in 2022, partially offset by lower R&D and MG&A spending, and a higher tax benefit.		Lower operating cash flow primarily driven by lower net income, partially offset by favorable changes in working capital and other adjustments.	

Managing to our long-term financial model

Our 2023 results reflect the continued advancement of our transformational journey. We continued to prioritize investments critical to our IDM 2.0 transformation, achieved operational milestones, and executed disciplined expense management. To achieve our long-term financial model, we believe it is imperative that we drive to world-class product cost and operational efficiency. A key component of our overall strategy is our internal foundry model. Under this model, we intend to reshape our operational dynamics and establish transparency and accountability through standalone profit and loss reporting for our manufacturing group in 2024. We expect this model to enable increased efficiencies across a number of aspects of our organization that we believe are integral to achieving our financial and operational goals. At the same time, we continue to prioritize capital investments critical to our efforts to regain process leadership and establish a leading-edge, at-scale foundry business.

¹ See "Non-GAAP Financial Measures" within MD&A.

Delivering leadership products

We seek to develop and offer leading products that will help enable a future in which every human can have more computing power and quicker access to data. We remain committed to our goal of delivering five technology nodes in four years to regain transistor performance and power performance leadership by 2025. This year, we achieved several key milestones on our product roadmap, including:

- We launched the Intel® Core™ Ultra processors, featuring our first integrated neural processing unit, for power-efficient AI acceleration and local inference on the PC.
- We introduced the 13th Gen Intel® Core™ mobile processor family, led by the launch of the first 24-core processor for laptops, and introduced the new Intel vPro® Platform powered by the full lineup of 13th Gen Intel Core processors.
- We introduced the 14th Gen Intel® Core™ desktop processor family, delivering fast desktop frequencies and enhanced desktop experience for enthusiasts.
- We launched the 4th Gen Intel® Xeon® Scalable processors, a critical part of our heterogeneous hardware and software portfolio to accelerate real-world data center, cloud and edge workloads, including AI, and also launched the 4th Gen Intel Xeon Scalable processors with Intel® vRAN Boost, a new general-purpose chip that fully integrates Layer 1 acceleration into the Xeon SoC and is designed to eliminate the need for external accelerator cards.
- We launched the 5th Gen Intel® Xeon® Scalable processors for data center, cloud, and edge, with embedded capabilities for powering AI workloads.
- We introduced two new Intel® Arc™ Pro Graphics Processing Units, Intel Arc Pro A60 and Intel Pro A60M, and shipped Intel Arc Pro A40-based systems.

Investing in leading-edge, at-scale manufacturing

We are committed to strengthening the resilience of the global semiconductor supply chain for leading-edge semiconductor products by investing in geographically balanced manufacturing capacity. In the US, we are expanding our existing operations in Arizona, New Mexico, and Oregon, and investing in two new leading-edge chip factories in Ohio. We have submitted all four of our major project proposals in Arizona, New Mexico, Ohio, and Oregon, estimated to represent over \$100 billion of US manufacturing and research investments over the next five years, to the US Department of Commerce's CHIPS Program Office.

In the European Union and Israel, we have announced a series of investments, spanning our existing operations in Ireland and Israel, as well as our planned investment of more than \$33 billion in Germany to build a leading-edge wafer fabrication mega-site, and our plans to invest up to \$4.6 billion in an assembly and test facility in Poland. We also announced the start of high-volume manufacturing using Intel 4 technology and EUV technology in Ireland.

Unlocking value

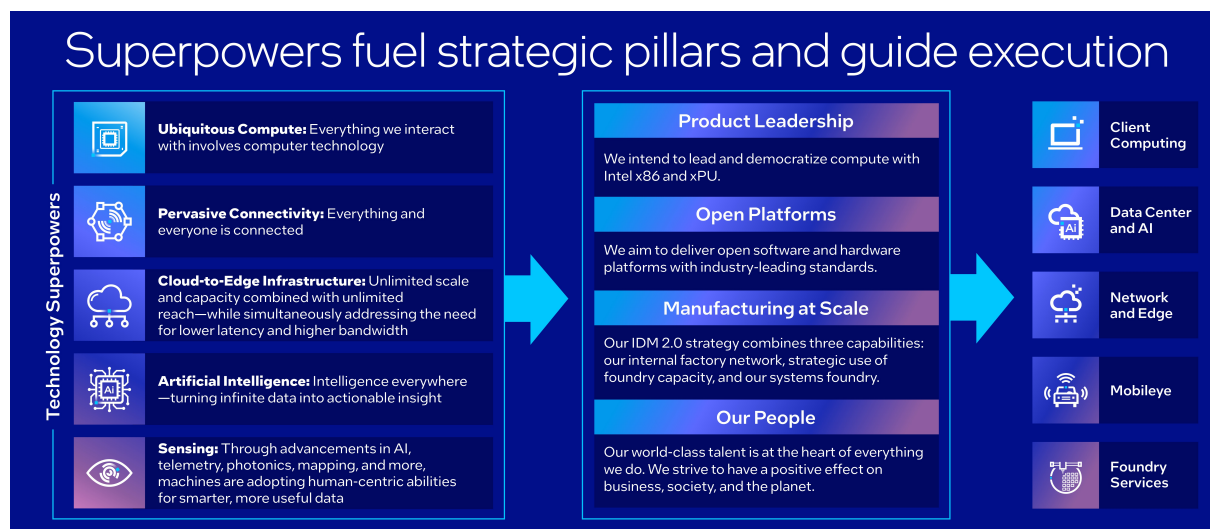
We continue to look for innovative ways to unlock value for our stakeholders. We sold a 32.4% minority stake in our IMS Nanofabrication business, including investments from Bain Capital and Taiwan Semiconductor Manufacturing Company (TSMC). Net proceeds resulting from the IMS minority stake sales totaled \$1.4 billion. We also executed a secondary offering of Mobileye stock that generated net proceeds of \$1.6 billion. Further, we communicated our intent to operate our Programmable Solutions Group (PSG) as a standalone business, with standalone financial reporting beginning January 1, 2024. This is expected to enable potential private and public equity investments. These transactions provide an additional source of capital to support the critical investments needed to advance our strategy.

Our Strategy

Technology permeates every aspect of our lives and is increasingly central to every aspect of human existence. As we look ahead to the next decade, we expect to see continued demand for processing power. Semiconductors are the underlying technology powering this digital expansion, and we are strategically positioning ourselves to create a resilient global semiconductor supply chain by investing in geographically balanced manufacturing capacity. The demand for compute is being accelerated by five superpowers: ubiquitous compute, pervasive connectivity, cloud-to-edge infrastructure, AI, and sensing. Together these superpowers combine to amplify and reinforce each other, and will exponentially increase the world's need for computing by packing even more processing capability onto ever-smaller microchips. We intend to lead the industry by harnessing these superpowers for our customers' growth and our own.

We are uniquely positioned with the depth and breadth of our silicon, platforms, and software, and packaging and process technology with at-scale manufacturing. With these strengths and the tailwinds of the superpowers driving digital disruption, our strategy to win is focused on four key themes: product leadership, open platforms, manufacturing at scale, and our people.

Our Priorities



Product Leadership

Lead and democratize compute with Intel x86 and xPU. Our product offerings provide end-to-end solutions, scaling from data center to network, PCs, edge computing, and the emerging fields of AI and autonomous driving, to serve an increasingly smart and connected world.

At our core is the x86 computing ecosystem, which supports an extensive and deep universe of software applications, with billions of lines of code written and optimized for x86 CPUs. We continue to advance this ecosystem with x86 microarchitectures focused on performance, which push the limits of low latency and single-threaded application performance, and microarchitectures focused on efficiency, which are designed for computing throughput efficiency to enable scalable multithreaded performance.

Beyond the CPU, we are delivering a growing family of xPU products, which encompass client and data center GPUs, IPUs, FPGAs, and other accelerators. The xPU approach recognizes that different workloads benefit from different computing architectures, and our broad portfolio helps meet our customers' increasingly diverse computing needs. As part of our strategy, we seek to develop and offer leading products across each of these architectural categories.

We also seek to address every phase of the AI continuum, including the largest, most challenging GenAI and large language models. We believe AI represents a generational shift in computing by expanding human abilities and solving the most challenging problems. We are in the early stages of realizing AI's full potential and GenAI is just the beginning. Our strategy is to bring AI to where the data is being generated and used and we believe we have a full spectrum of hardware and software platforms, offering open and modular solutions, for competitive total cost of ownership and time to value that customers need to win in this era of exponential growth and AI everywhere. We are infusing AI into Intel technologies, supporting today's GenAI workloads, fueling emerging usages like AI PC and AI at the edge, and pioneering innovations that we believe advance the future of AI in the next decade. We believe our leadership in IP, process, packaging, security, software, services, manufacturing, and foundry services positions us to realize AI's full potential to transform industries and solve the world's biggest challenges.

Open Platforms

We aim to deliver open software and hardware platforms with industry-defining standards. Around the globe, companies are building their networks, systems, and solutions on open standards-based platforms. Intel has helped set the stage for this movement, with our historic contributions in developing standards such as CXL, Thunderbolt™, and PCI Express*. We also contributed to the design, build, and validation of open-source products in the industry such as Linux*, Android*, and others. The world's developers constantly innovate and expand the capabilities of these open platforms while increasing their stability, reliability, and security. In addition, microservices have enabled the development of flexible, loosely coupled services that are connected via application programming interfaces to create end-to-end processes. We use industry collaboration, co-engineering, and open-source contributions to accelerate software innovation. Through our oneAPI initiative, developers use a unified language across CPUs, GPUs, and FPGAs that is designed to reduce development time and to enhance productivity. We also deliver a steady stream of open-source code and optimizations that are designed for projects across virtually every platform and usage model. We are committed to co-engineering and jointly designing, building, and validating new products with software industry leaders to accelerate mutual technology advancements and help new software and hardware work better together. Our commitment extends to developers through our developer-first approach based on openness, choice, and trust.

Ultimately, we believe our pivot to a software-defined, silicon-enhanced strategy will enable us to realize value at all layers of the compute stack. This should allow us to continue to monetize foundational and ecosystem-enabling software through hardware sales, limited licensing, and customer-enabling service offerings. Additionally, we are expanding our software portfolio by developing and monetizing software solutions, services, and platforms with SaaS, software subscriptions, and other business models. We are prioritizing three portfolios of offerings for our SaaS and subscription-based software: AI, trust and security, and performance optimization. We are also scaling the availability of Intel® Developer Cloud, which is designed to enable developers to learn, prototype, test, and run their own AI workloads across multiple Intel hardware architectures to experience the competitive performance of Intel platforms and develop their AI software from today's hardware portfolio to next-generation architectures.

An open approach and deep engagement with the developer ecosystem are essential to lowering barriers to entry and unlocking AI innovations for developers and customers. We are expediting an open AI software ecosystem that we believe is needed to break down proprietary walls. We offer customers, partners, and developers early access and a rapid path to scale their AI solutions with the Intel Developer Cloud and integrated and scalable hardware and software systems and solutions.

We believe AI will only be truly accessible to all when its use is ethical and responsible. Partnering with industry leaders, we are working to deliver innovative ecosystem tools and solutions intended to make AI safer and more secure, and help address privacy concerns as AI scales exponentially. We are building platforms and technologies for the convergence of AI and security to help customers confidently secure diverse AI workloads across the data center, cloud, PC, and edge.

Manufacturing at Scale

IDM 2.0, the next evolution and expansion of our IDM model, is a differentiated strategy that combines three capabilities:

Internal factory network. Our global, internal factory network has been foundational to our success, enabling product optimization, improved economics, and supply resilience. We intend to remain a leading developer of process technology and a major manufacturer of semiconductors and will continue to build the majority of our products in our factories.

Strategic use of foundry capacity. We expect to expand our use of third-party foundry manufacturing capacity, which will provide us with increased flexibility and scale to optimize our product roadmaps for cost, performance, schedule, and supply. Our use of foundry capacity will include manufacturing for a range of modular tiles on advanced process technologies.

Open System Foundry. We are building a world-class foundry business to meet the growing long-term global demand for semiconductors. We plan to differentiate our foundry offerings from those of others through a combination of leading-edge packaging and process technology, committed capacity in the US and Europe available for customers globally, and a world-class IP portfolio that will include x86 cores, as well as other ecosystem IP. The current foundry model enabled explosion of ecosystem innovation at the wafer level. We believe this established model has historically served the industry well, but a new mindset is needed in our new era of chipmaking. As innovation evolves, we see the rack has collapsed into a system and the system has collapsed into an advanced package. We are building out an Open System Foundry that has four components: wafer fabrication, packaging, chiplet standard, and software.

The Open System Foundry involves engaging with customers at multiple levels, from basic wafer manufacturing to helping define and implement their desired system architectures. We intend to build our customers' silicon designs and deliver full end-to-end customizable products built with our advanced packaging technology.

We believe our IDM 2.0 strategy enables us to deliver leading process technology and products to meet growing long-term demand using internal and external capacity, while leveraging our core strengths to provide foundry services to others and providing superior capacity, supply resilience, and an advantageous cost structure.

In the upcoming year, we plan to implement an internal foundry model, where our business units engage with our manufacturing group in an arm's-length fashion, similar to how fabless semiconductor companies engage with external foundries. The model is integral to our IDM 2.0 operational and financial strategies and aims to fully leverage our invested capital while serving a wide variety of chip customers worldwide.

Recent events, including the pandemic-related global chip shortage, made clear how supply chain disruptions can severely impact everyday life. The steady rise of chipmaking capacity in Asia has made the world vulnerable to continued and increasingly extreme shortages. We believe a secure, balanced, and resilient supply of semiconductors is essential to the interests of the entire global economy. With one of the most geographically balanced supply chains across North America, Europe and Asia, we plan to partner with the public sector to strengthen the resilience of the global semiconductor supply chain by investing in a more geographically balanced capacity. We believe our investments will help develop a more resilient supply chain for generations to come and hedge against geopolitical instability.

Our People

[Our world-class talent is at the heart of everything we do. Together we strive to have a positive effect on business, society, and the planet.](#) Delivering on our strategy and growth ambitions requires attracting, developing, and retaining top talent from across the world. Our people build our technology, unlock new business opportunities, and work with our partners and customers to create global impact.

Fostering a culture of empowerment, inclusion, and accountability is also core to our strategy. We are committed to creating an inclusive workplace where the world's best engineers and technologists can fulfill their dreams and create technology that improves the life of every person on the planet.

Growth Imperative

We are investing to position the company for accelerated long-term growth, focusing on both our core businesses and our growth businesses. In our client and server businesses, our strategy is to invest to strengthen the competitiveness of our product roadmap and to explore new opportunities. We believe we have significant opportunities to grow and gain share in graphics; mobility, including autonomous driving; networking and edge; AI; software; and foundry services.

Focus on Innovation and Execution

We are focused on executing our product and process roadmap and accelerating our cadence of innovation. We have set a detailed process and packaging technology roadmap and announced key architectural innovations to further our goal of delivering leadership products in every area in which we compete. We are returning our culture to its roots in innovation and execution, drawing on principles established by our former CEO Andy Grove that emphasize discipline and accountability. This includes using OKRs throughout the organization to drive a common purpose.

To help us execute toward our IDM 2.0 strategy, we are leveraging our Smart Capital approach. This approach is designed to enable us to adjust quickly to opportunities in the market, while managing our margin structure and capital spending. The key elements of Smart Capital include:

- [Smart capacity investments](#). We are aggressively building out manufacturing shell space, which gives us flexibility in how and when we bring additional capacity online based on milestone triggers such as product readiness, market conditions, and customer commitments.
- [Government incentives](#). We are continuing to work with governments in the US and Europe to advance and benefit from incentives for domestic manufacturing capacity for leading-edge semiconductors.
- [SCIP](#). We are accessing strategically aligned capital to increase our flexibility and help efficiently accelerate and scale manufacturing build-outs. This type of co-investment also demonstrates how private capital is unlocked and becomes a force multiplier for government incentives for semiconductor manufacturing expansion.
- [Customer commitments](#). IFS is working closely with potential customers to make advance payments to secure capacity. This provides us with the advantage of committed volume, de-risking investments while providing capacity corridors for our foundry customers.
- [External foundries](#). We intend to continue our use of external foundries where their unique capabilities support our leadership products.

Our Capital

We deploy various forms of capital to execute our strategy in a way that seeks to reflect our corporate values, help our customers succeed, and create value for our stakeholders.

Capital	Strategy	Value
Financial		
	Leverage financial capital to invest in ourselves and exit businesses to optimize our portfolio, both to drive our strategy and long-term value creation.	We strategically invest financial capital to continue to build our business, create long-term value and provide returns to our stockholders.
Intellectual		
	Invest significantly in R&D and IP to enable us to deliver on our accelerated process technology roadmap, introduce leading x86 and xPU products, and develop new businesses and capabilities.	We develop IP to enable next-generation products, create synergies across our businesses, expand into new markets, and establish and support our brands.
Manufacturing		
	Build manufacturing capacity efficiently to meet the growing long-term global demand for semiconductors, aligned with our IDM 2.0 strategy.	Our geographically balanced manufacturing scope and scale enable us to provide our customers with a broad range of leading-edge products and foundry capabilities.
Human		
	Build a diverse, inclusive, and safe work environment to attract, develop, and retain top talent needed to build transformative products.	Our talented employees enable the development of solutions and enhance the intellectual and manufacturing capital critical to helping our customers win the technology inflections of the future.
Social and Relationship		
	Build trusted relationships for both Intel and our stakeholders, including employees, suppliers, customers, local communities, and governments.	We collaborate with stakeholders on programs to empower underserved communities through education and technology, and on initiatives to advance accountability and capabilities across our global supply chain, including accountability for the respect of human rights.
Natural		
	Strive to reduce our environmental footprint through efficient and responsible use of natural resources and materials used to create our products.	With our proactive efforts, we seek to mitigate climate and water impacts, achieve efficiencies, lower costs, and position ourselves to respond to the expectations of our stakeholders.

Comprehensive ESG and Corporate Responsibility Strategy: RISE

Our commitment to corporate responsibility and sustainability leadership is deeply integrated throughout our business. We strive to create an inclusive and positive work environment where every employee has a voice and a sense of belonging, and we are proactive in our efforts to reduce our environmental footprint through efficient and responsible use of natural resources and materials.

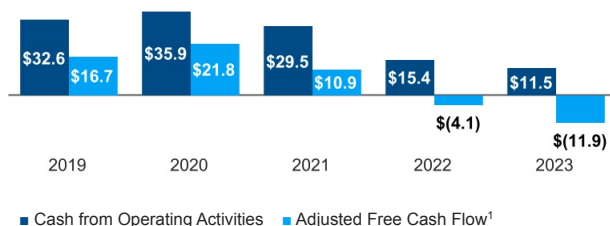
We continue to raise the bar for ourselves and leverage our leadership position in the global technology ecosystem to make greater strides in corporate responsibility and apply technology to address social and environmental challenges. Through our **RISE** strategy, we aim to create a more **responsible**, **inclusive**, and **sustainable** world, **enabled** by our technology and the expertise and passion of our employees. Our corporate responsibility strategy is designed to increase the scale of our work through collaboration with our stakeholders and other organizations; we know that we cannot achieve the broad social impact to which we aspire by acting alone. More information about our RISE goals, including progress we have made toward achieving them, is included in our Corporate Responsibility Report¹.

¹ The contents of our Corporate Responsibility Report are referenced for general information only and are not incorporated by reference in this Form 10-K.

Financial Capital

We take a disciplined approach to our financial capital allocation strategy, which continues to focus on building stakeholder value and is driven by our priority to invest in the business and capacity and our capital needs. We also seek to pay competitive dividends, optimize our portfolio, look for innovative ways to unlock value across our assets, and, from time-to-time, engage in mergers and acquisitions. As we invest in our IDM 2.0 strategy and implement our next phase of capacity expansions and the acceleration of our process technology roadmap, our allocation priorities have shifted more heavily toward investing in the business and away from stock repurchases. In the long term, we will continue to look for opportunities to further our strategy through acquisitions while remaining disciplined on capital allocation.

Cash from Operating Activities \$B



Our Financial Capital Allocation Strategy

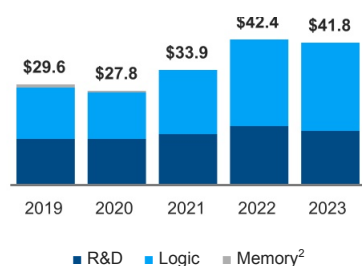
Invest in the Business

Our first allocation priority is to invest in R&D and capital spending to capitalize on the opportunity presented by the world's demand for semiconductors. In 2023, we continued our focus on capital investment and the deployment of our Smart Capital strategy.

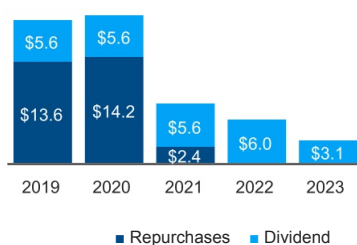
Return Excess Cash to Stockholders

Our capital allocation strategy includes returning excess cash to stockholders. We achieve this through our dividend policy and, when permissible, stock repurchases. In 2023, we declared a reduced quarterly dividend on our common stock. This dividend reduction reflects our deliberate approach to capital allocation, is expected to support the critical investments needed to execute our business strategy, and is designed to position us to create long-term value. We expect future stock repurchases to continue to be curtailed during this time of meaningful investment in capital.

R&D and Capital Investments \$B



Cash to Stockholders \$B



Optimize our Portfolio and Unlock Value

Our capital allocation strategy also includes opportunistic investment in and acquisition of companies that complement our strategic objectives. We look for acquisitions that supplement and strengthen our capital and R&D investments.

We also seek to drive value creation through transactions such as the 2022 Mobileye IPO, the 2023 minority stake sales in IMS, and the 2023 announcement of our intent to operate PSG as a standalone business which we expect to enable potential private and public equity investments. Together, these transactions provide an additional source of capital to support the critical investments needed to advance our business strategy.

Lastly, we take action when investments do not strategically align to our key priorities. In the last three years, we exited numerous businesses, including the NAND memory business and Intel® Optane™ memory business.

¹ See "Non-GAAP Financial Measures" within MD&A.

² 2021-2023 capital investments in Memory are not presented due to the divestiture of the NAND memory business announced in October 2020. 2019 and 2020 capital investments presented include Memory.



Intellectual Capital

Research and Development

R&D investment is critical to enable us to deliver on our accelerated technology roadmap, introduce leading products, and develop new businesses and capabilities in the future. We seek to protect our R&D efforts through our IP rights and may augment R&D initiatives by acquiring or investing in companies, entering into R&D agreements, and directly purchasing or licensing technology.

Areas Key to Product Leadership

We have intensified our focus on areas key to product leadership. Our objective with each new generation of products is to improve user experiences and value through advances in performance, power, cost, connectivity, security, form factor, and other features. We also focus on reducing our design complexity, re-using IP, and increasing ecosystem collaboration to improve our efficiency.

Process and packaging. Our leading-edge process and packaging technology and world-class IP portfolio are key to the success of our strategy. We are committed to achieving process technology leadership in 2025 by planning to deliver five technology nodes in four years. In addition, we have solidified our process and packaging offerings to external customers through IFS.

- Intel 7 process node is in production for our 13th Gen Intel Core processors and represents the continuous improvement of transistor and interconnect performance.
- Intel 4, our first EUV node, delivers significant density scaling and approximately 20% performance-per-watt improvement over Intel 7. The Intel Core Ultra processor is our first high-volume client product on Intel 4 and began shipping to customers in Q3 2023.
- Intel 3 is expected to deliver further logic scaling and up to 18% performance-per-watt improvement over Intel 4. Intel 3 is our first advanced node offered to IFS customers and is optimized for the needs of data center products.
- Intel 20A will follow Intel 3 and will introduce two breakthrough technologies that we expect to deliver up to a 15% performance-per-watt improvement over Intel 3: RibbonFET and PowerVia. RibbonFET, our implementation of a gate-all-around transistor, is designed to deliver faster transistor switching speeds while achieving the same drive current as multiple fins, but in a smaller footprint. PowerVia is our unique industry-first implementation of backside power delivery that is designed to optimize signal transmission by eliminating the need for power routing on the front side of the wafer.
- Intel 18A, our second IFS advanced node offering, will improve on Intel 20A by delivering ribbon innovation for design optimization and line width reduction. Intel 18A is expected to deliver an additional 10% improvement in performance per watt over Intel 20A.
- Beyond Intel 18A, we have initiated definition and development of our next two process nodes and continue to define, build, and develop the next-generation High Numerical Aperture EUV lithography into our process technology roadmap. To continue our modernization and infrastructure expansions, we have begun the installation of the world's first High-NA EUV tool for commercial use at our Gordon Moore Park in Oregon.
- Our family of 3D advanced packaging technology will usher in the next generation of Foveros technology, enabling us to mix multiple top die tiles with multiple base tiles across mixed fab nodes, providing greater flexibility for disaggregated chip designs. Our future Foveros Direct technology should scale interconnect pitch below 10µm, enable direct copper-to-copper bonding for low-resistance interconnects, and blur the boundary between wafer and package. Longer term, we expect to deliver complete glass substrate solutions as the next generation of advanced packaging designed to enable continued scaling of transistors in a package.

xPU architecture. We believe the future is a diverse mix of scalar, vector, matrix, and spatial architectures deployed in CPU, GPU, IPU, accelerator, and FPGA sockets, enabled by a scalable software stack and integrated into systems by advanced packaging technology. We are building processors that span several major computing architectures, moving toward an era of heterogeneous computing:

- **CPU.** We started shipping our 5th Gen Intel Xeon Scalable processors based on Intel 7 with built-in AI acceleration, cryptographic acceleration, and advanced security capabilities. We also launched our 14th Gen Intel Core processors, which are designed to scale from thin and light laptops to enthusiast desktop and notebook platforms. These are based on a hybrid architecture utilizing our most advanced performance cores and power-efficient cores.
- **GPU and HPC.** The Intel Arc graphics family offers modern GPU features to power today's demanding games, applications, and AI workloads. In 2023, the Intel Arc portfolio expanded while offering a substantial performance uplift through continual driver updates, offering consumer and professional solutions that target a significant part of the market. Intel Arc discrete GPUs currently ship in three performance tiers, with Intel Arc 7, 5, and 3 series for consumers, and Intel Arc Pro A60, A50, and A40 for workstations. In the mobile segment, Intel Arc A-series GPUs can be found powering a range of consumer and workstation designs. In Q1 2023, we launched the Intel® Xeon® CPU Max Series and Intel® Data Center GPU Max Series. Designed for the future compute demands of HPC and AI workloads, the Intel Max Series product family power the Aurora supercomputer at Argonne National Laboratory.

- **Infrastructure Acceleration.** Our first ASIC IPU has been deployed widely across Google Cloud. Our second generation IPU, is also co-developed with Google and is expected to tape in the first half of 2024. It is expected to double the throughput and compute capabilities. The IPU becomes the center of the CSP compute platform, with the bulk of CSP infrastructure cycles spent running on the IPU and not the host processor. This frees up processing cores for business-value applications and infrastructure as a service rental.
- **Ethernet.** Our 200G foundational network interface card targeting cloud, enterprise, and edge, is expected to go into production in 2H 2024 and extend our 100G market. Additionally, we plan to productize our 10G/2.5G/1G offering, a refresh of our high-volume 10G and 1G portfolio with modernized software, management, and manufacturing technologies.
- **Matrix Accelerator.** We are showing strong customer and performance momentum for our Intel® Gaudi® AI accelerators. In Q3 2023, we announced that a large AI supercomputer would be built entirely on Intel Xeon processors and 4,000 Intel Gaudi 2 AI hardware accelerators, with Stability AI as the anchor customer.

Software. Software unleashes the potential of our hardware platforms across all workloads, domains, and architectures.

- oneAPI adoption continues to expand across the industry. oneAPI is designed to enable developers to build cross-architecture applications using a single code base across CPUs, GPUs, and FPGAs to reduce development time and enhance productivity. Our oneAPI-based tools take advantage of unique hardware features and lower software development and maintenance costs. Developers can choose the best architecture for the problem at hand without rewriting their entire code base, accelerating their time to value.
- We seek to accelerate adoption of oneAPI and Intel software developer tools through diverse ecosystem activities, including developer training, summits, centers of excellence, access to Intel hardware and software through Intel Developer Cloud and industry collaboration through the Linux Foundation's Unified Acceleration (UXL) Foundation project. Our Intel Developer Cloud offering is designed to host global users spanning AI, data science, high-performance computing, media and graphics, and other accelerated computing workloads using oneAPI and the latest Intel hardware.
- We believe AI will be ubiquitous, and with our tools and the broad open software ecosystem, we are well-positioned to scale AI. We optimize and contribute software to popular frameworks, such as PyTorch and TensorFlow, to achieve the best performance for Intel platforms and to deliver productivity and programmability for AI developers. We are accelerating the disruption of the proprietary ecosystems through Intel contributions to new software AI languages such as Triton, Mojo, and JAX. This higher level of open, vendor-independent programmability, combined with Intel's commitment to open spec governance, implementations, licensing models, and segment-optimized solutions, such as OpenVINO™ powered by oneAPI, offers customers a choice of hardware and software stack tailored to diverse workload needs.
- We seek to continually improve our system and foundational-level software in support of our client, data center, networking, and graphics products, delivering AI-optimized software across the stack, including BIOS, firmware, simulation, operating systems, and virtualization.

IP Rights

We own and develop significant IP and related IP rights around the world that support our products, services, R&D, and other activities and assets. Our IP portfolio includes patents, copyrights, trade secrets, trademarks, mask works, and other rights. We actively seek to protect our global IP rights and deter unauthorized use of our IP and other assets.

We have obtained patents in the US and other countries. Because of the fast pace of innovation and product development, our products are often obsolete before the patents related to them expire, and in some cases our products may be obsolete before the patents are granted. As we expand our product offerings, particularly around our foundry business, we also seek to extend our patent development efforts. In addition to developing patents based on our own R&D efforts, we may purchase or license patents from third parties.

The software that we distribute, including software embedded in our products, is entitled to copyright and other IP protection. To distinguish our products from our competitors' products, we have obtained trademarks and trade names for our products, and we maintain cooperative advertising programs with customers to promote our brands and to identify products containing genuine Intel components. We also protect details about our processes, products, and strategies as trade secrets, keeping confidential the information that we believe provides us with a competitive advantage.

Efforts to protect our IP can be difficult, particularly in countries that provide less protection to IP rights and in the absence of harmonized international IP standards. Competitors and others may already have IP rights covering similar products. There is no assurance that we will be able to obtain IP rights covering our own products or that we will be able to obtain IP licenses from other companies on favorable terms or at all. For a discussion of IP-related risks, see "Risk Factors" within Risk Factors and Other Key Information. While our IP rights are important to our success, our business as a whole is not significantly dependent on any single patent, copyright, or other IP right.



Manufacturing Capital

As the guardians of Moore's Law, we continue to innovate to advance the design and manufacturing of semiconductors to help address our customers' greatest challenges. This makes possible new leadership products with higher performance while balancing power efficiency, cost, and size.

Our IDM 2.0 strategy allows us to deliver leadership products using internal and external capacity while leveraging our core strengths to provide foundry services to others. IDM 2.0 combines three capabilities. First, we will continue to build most of our products in our fabs. Second, we expect to expand our use of third-party foundry capacity to manufacture a range of modular tiles on advanced process technologies. Third, we are building a world-class foundry business with IFS, which we expect will combine leading-edge packaging and process technology, committed capacity in the US and Europe, and a world-class IP portfolio that will include x86 cores, as well as other ecosystem IP.

Network and Supply Chain

In 2023, our factories delivered continuous support to our customers as we ramped new process technologies and equipment for our products and expanded Open System Foundry offerings. We continue to work across our supply chain to minimize disruptions, improve productivity, and increase overall capacity and output to meet customer expectations.

Our global supply chain supports internal partners across architecture, product design, technology development, manufacturing and operations, sales and marketing, and business units, and our supply ecosystem comprises thousands of suppliers globally. Our mission is to enable product and process leadership, industry-leading total cost of ownership, and on-time and uninterrupted supply for our customers, delivered in a responsible and sustainable manner.

As of the end of 2023, we had nine geographically dispersed manufacturing sites in production. The following map shows the locations of these factory sites. Our manufacturing facilities are primarily used for silicon wafer manufacturing, assembling, testing, and advanced packaging. We operate in a network of manufacturing facilities integrated as though they were one factory to provide the most flexible supply capacity, allowing us to better analyze our production costs and adapt to changes in capacity needs. Our new process technologies, when ready for high volume manufacturing, are transferred from a central development fab to one or more of our manufacturing facilities. The network of factories and the development fab collaborate to continue driving operational improvements. This enables fast ramp of the operation, quick learning, and quality control.

We are expanding manufacturing capacity across multiple sites and geographies. These include silicon wafer manufacturing in Arizona, Germany, Ireland, Israel, Ohio, and Oregon and advanced packaging manufacturing in Malaysia and New Mexico. This year, we added Poland to our assembly and test expansion roadmap. These investments further our IDM 2.0 strategy and are expected to support a resilient semiconductor supply chain and to create the foundation for a next-generation chip ecosystem.

Intel's Global Footprint



Human Capital

Our human capital strategy is grounded in our belief that our people are fundamental to our success. Delivering on our strategy and growth ambitions requires attracting, developing, and retaining top talent across the world. We are committed to creating an inclusive workplace where the world's best engineers and technologists can fulfill their dreams and create technology that improves the life of every person on the planet. We invest in our highly skilled workforce of 124,800 people (as of December 30, 2023) by creating practices, programs, and benefits that support the evolving world of work and our employees' needs.

Our human capital philosophy includes three pillars we believe are needed to position our talent strategy as a competitive advantage:

- **Hire and retain the best talent:** We have embraced the future of work with a flexible, hybrid-first approach that differentiates Intel from competitors.
- **Develop our talent to full potential:** We have updated role descriptions and career pathways to better enable movement and help top talent work on the highest priorities.
- **Create a winning culture:** We have reignited Intel's results-driven, performance culture.

Fostering a culture of empowerment, inclusion, and accountability is also core to our strategy. We believe that an inclusive culture is important for attracting, developing, and retaining top talent, and we strive to provide a work environment where all employees from all backgrounds are valued, challenged, and rewarded. We are focused on reinvigorating our culture to strengthen our execution and accelerate our cadence of innovation. Our values—customer first, fearless innovation, results driven, one Intel, inclusion, quality, and integrity—inspire us and are key to delivering on our purpose. All employees are responsible for upholding these values, the Intel Code of Conduct, and Intel's Global Human Rights Principles, which form the foundation of our policies and practices and ethical business culture.

Talent Management

We continue to see significant competition for talent throughout the semiconductor industry. Our hiring was limited in 2023, in line with macroeconomic forecasts, financial performance, and cost-cutting measures, and we took actions to rebalance our workforce. However, the investments we are making to accelerate our process technology require continued and focused efforts to attract and retain talent—especially technical talent. Our undesired turnover rate¹ was 5.6% in both 2023 and 2022.

We invest significant resources to develop the talent needed to remain at the forefront of innovation and make Intel an employer of choice. We offer extensive training programs and provide rotational assignment opportunities and have updated our job architecture to help employees create custom learning curricula for building skills and owning their careers. To further support the growth and development of our people, we continue to increase mentoring in our technical community, drive engagement through employee resource groups, and promote health and wellness resources to all our people. Through our annual employee experience survey, employee inclusion survey, and manager development feedback survey, employees can voice their perceptions of the company, their managers, their work experiences, and their learning and development opportunities. Our employees' voices are important to enable our culture of continuous improvement, and as a result, we link a portion of our executive and employee performance bonus to year-over-year improvements of our employee experience survey results. Our performance management system is designed to support our cultural evolution and to increase our focus on disciplined OKRs.

¹ Undesired turnover includes all regular Intel employees who voluntarily left Intel, but does not include Intel contract employees, interns, or employees who separated from Intel due to divestiture, retirement, voluntary separation packages, death, job elimination, or redeployment.

Inclusion

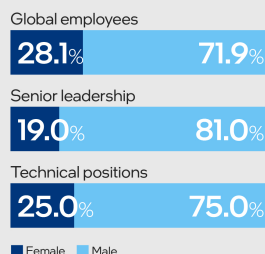
Diversity and inclusion are core elements of Intel's values and instrumental to driving innovation and positioning us for growth. Over the past decade, we have taken actions to integrate diversity and inclusion expectations into our culture, performance and management systems, leadership expectations, and annual bonus metrics. Through our annual employee inclusion survey, employees can voice their experiences at Intel and provide feedback on how we can continue to improve. We are proud of what we have accomplished to advance diversity and inclusion, but we believe we can achieve more, including beyond the walls of Intel. Our RISE strategy and 2030 goals set our global ambitions for the rest of the decade, including, in part, ambitious goals to achieve 25% representation of women in senior leadership; achieve 12% representation of underrepresented minorities in US senior leadership; achieve 10% representation of employees with a disability in our global workforce; and exceed 40% representation of women in technical roles, including engineering positions and other roles with technical job requirements. Our ambitious goals were established, in part, based on modeling of where market availability is expected to be in 2030. Market availability analysis will be re-conducted over the time period to assess ongoing alignment and expectations. To drive accountability, we continue to link a portion of our executive and employee compensation to diversity and inclusion metrics.

We have committed our scale, expertise, and reach through our comprehensive RISE strategy to work with customers and other stakeholders to accelerate the adoption of inclusive business practices across industries. As part of the Alliance for Global Inclusion, we worked with a coalition of technology companies to create a global inclusion index survey, which serves as a benchmark for companies to track diversity and inclusion improvements, provide information on current best practices, and highlight opportunities to improve outcomes across industries. The results of the third global inclusion index survey were published in 2023 and shared with business leaders across industries. For the first time, based on the maturity of the respondents' best practices, the survey results transitioned to an index, in which organizations that received an overall score of 50% or higher across all regions in which they participated were recognized for their diversity and inclusion efforts. There were 27 respondents and, of those, 18 earned spots on the index, including Intel. This collective effort allows the industry to more clearly identify actions needed to advance progress on closing persistent gaps and advancing more inclusive practices in workplaces, industry, and society. The survey shows increased commitment to diversity and inclusion around key areas of workforce inclusivity, recruitment, advancement opportunities, and accountability. We will also continue to collaborate on initiatives that expand the diverse pipeline of talent for our industry, advance social equity, make technology fully inclusive, and expand digital readiness for millions of people around the world.

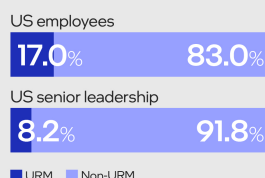
² Senior leadership refers to salary grades 10+ and equivalent grades. While we present male and female, we acknowledge this is not fully encompassing of all gender identities.

³ The term underrepresented minority (URM) is used to describe diverse populations, including Black/African American, Hispanic, and Native American employees in the US.

2023 Global Employees by Gender²



2023 URM³ in the US



Compensation and Benefits

We structure pay, benefits, and services to meet the varying needs of our employees, helping support employee financial well-being with competitive compensation, investment opportunities, and financial resources. Our total rewards package includes market-competitive pay, broad-based stock grants and bonuses, an employee stock purchase plan, healthcare and retirement benefits, paid time off and family leave, parent reintegration, fertility assistance, flexible work schedules, sabbaticals, and on-site services. Since 2019, we have achieved gender pay equity globally and we continue to maintain race/ethnicity pay equity in the US. We achieve pay equity by closing the gap in average pay between employees of different genders or race/ethnicity in the same or similar roles after accounting for legitimate business factors that can explain differences, such as location, time at grade level, and tenure. We have also advanced transparency in our pay and representation data by publicly releasing our EEO-1 survey pay data since 2019. We believe that our holistic approach toward pay equity, representation, and creating an inclusive culture enables us to cultivate a workplace that helps employees develop and progress in their careers at all levels. Our "hybrid-first" approach to working was informed by employees surveyed around the globe and involves the majority of our employees splitting their time between working remotely and in the office. Hybrid-first and remote work options cast a wider recruitment net and support our ambition to hire the best global talent. Currently, there is no company-wide mandate on the number of days per week employees should be on site or how they should collaborate. Our goal is to enable remote and on-site work where it drives the best output, while providing our employees with equitable access to systems, resources, and opportunities that allow them to succeed.

Health, Safety, and Wellness

We are committed to providing a safe and injury-free workplace. We regularly invest in programs designed to improve physical, mental, and social well-being. We provide access to a variety of innovative, flexible, and convenient health and wellness programs, including on-site health centers, and we aim to increase awareness of and support for mental and behavioral health. In support of our RISE goals, we intend to continue our efforts to build our strong safety culture and drive the global expansion of our corporate wellness program through employee education and engagement activities.



Social and Relationship Capital

We are committed to engaging in initiatives that support our communities and help us develop trusted relationships with our stakeholders. Proactive engagement with our stakeholders and investments in social impact initiatives, including those aligned with the United Nations Sustainable Development Goals, advance our position as a leading corporate citizen and create shared value for Intel, our global supply chain, and our communities.

Economic and social. The health of our business and local economies depends in part on continued investments in innovation. We provide high-skill, high-paying jobs around the world, many of which are manufacturing and R&D jobs located in our factories. As we expand operations in both existing and new locations around the world, we are building a pipeline of qualified workers through our talent strategy and the many investments we are making in education. We also benefit economies through our R&D ecosystem spending, sourcing activities, employee spending, and tax payments. We make sizable capital investments and provide leadership in public-private partnerships to spur economic growth and innovation.

We stand at the forefront of new technologies that are increasingly being used to empower individuals, companies, and governments around the world to solve global challenges. We aim to empower people through education and advance social initiatives to create career pathways into the technology industry. This includes our global Intel Digital Readiness Programs, such as AI for Youth and AI for Workforce, scaled in partnership with governments and institutions to empower individuals with digital readiness and AI skills. Additionally, we invest in multi-year partnerships with historically Black colleges and universities in the US to increase the number of Black/African Americans who pursue electrical engineering, computer engineering, and computer science fields. Our employees and retirees share their expertise through volunteer initiatives in the communities where we operate, volunteering 3.8 million¹ hours over the past four years. These efforts contribute to our RISE goal to volunteer 10 million hours over a decade. Since 2020, we announced and further expanded upon the Intel RISE Technology Initiative, which provides an expanded channel to build deeper relationships with our customers and partners aligned with our corporate purpose and work to create shared value through our RISE strategy. Specifically, we are funding projects in areas such as using technology to improve health and safety, making technology more inclusive while expanding digital readiness, and carbon-neutral computing to help address climate change.

Human rights commitment. We are committed to maintaining and improving systems and processes to avoid causing or contributing to adverse impacts on human rights in our own operations, products, and supply chain. We have established an integrated approach to managing human rights across our business, including senior-level management involvement and board-level oversight. We also meet throughout the year with external stakeholders and experts on human rights to continue to inform and evolve our human rights policies and oversight processes. While we do not always know nor can we control what products our customers create or the applications end users may develop, we do not support or tolerate our products being used to adversely impact human rights. Where we become aware of a concern that Intel products are being used by a business partner in connection with abuses of human rights, we intend to evaluate and restrict or cease business with the third party unless and until we have a high confidence that Intel's products are not being used to adversely impact human rights.

¹ This is a preliminary estimate. The final number will be reported in our 2023-24 Corporate Responsibility Report, to be issued later in 2024.

Supply Chain Responsibility

We actively manage our supply chain to help reduce risk, improve product quality, achieve environmental and social goals, and improve overall performance and value creation for Intel, our customers, and our suppliers. To drive responsible and sustainable practices throughout our supply chain, we have robust programs to educate and engage suppliers that support our global manufacturing operations. We actively collaborate with other companies and lead industry initiatives on key issues such as improving transparency around climate and water impacts in the global electronics supply chain and, as part of our RISE strategy, we are advancing collaboration across our industry on responsible minerals sourcing. Through these efforts we help set electronics industry-wide standards, develop audit processes, and conduct training.

Over the past decade, we have directly engaged with suppliers to verify compliance and build capacity to address risks of forced and bonded labor and other human rights issues. We perform periodic audits, and identify critical direct suppliers to engage through capability-building programs, which help suppliers build sustainability acumen and verify compliance with the Responsible Business Alliance and the Intel Code of Conduct. We also engage with indirect suppliers through our programs on forced and bonded labor, responsible minerals, and supplier diversity. To achieve our RISE goals, we are significantly expanding the number of suppliers covered by our engagement activities. The supply chain environmental team is also actively engaging suppliers to measure and reduce their greenhouse gas emissions footprints and the resulting impact on our footprint. These activities are intended to help us meet our long-term emissions reductions goals, including our goal to achieve net-zero upstream Scope 3 greenhouse gas emissions by 2050, which we announced in 2023.

Our commitment to diversity and inclusion also extends to our suppliers. We believe a diverse supply chain supports greater innovation and value for our business. We have set additional spending targets with women-owned suppliers outside the US and with minority-owned suppliers globally to accelerate progress toward our goal to increase global annual spending with diverse suppliers to reach \$2.0 billion in annual spending by 2030.



Driving to the lowest possible environmental footprint as we grow helps us create efficiencies, support our communities, and respond to the needs of our stakeholders. We invest in environmental projects and set company-wide environmental targets to drive reductions in greenhouse gas emissions, energy and water use, and waste generation. We build energy efficiency into our products to help our customers lower their own emissions, energy usage, and costs, and we collaborate with policymakers and other stakeholders to use technology to address environmental challenges.

In April 2022, we announced our new goal to reach net-zero greenhouse gas emissions in our operations by 2040, creating an important target to strengthen our commitment to sustainable business practices. We furthered our commitment in 2023 by announcing our goal to reach net-zero upstream Scope 3 greenhouse gas emissions by 2050. Our 2030 RISE goals continue to be important milestones to drive to higher levels of operational efficiency, including a goal of a 10% reduction in our greenhouse gas emissions on an absolute basis from a 2019 baseline by 2030. We continue to take action on emissions reduction strategies focused on emissions abatement, and to make additional investments in renewable electricity, process and equipment optimization, and energy conservation. In 2023, we linked a portion of the executive and employee performance bonus to our goal to reduce our 2023 Scope 1 and 2 greenhouse gas emissions by 130,000 metric tons carbon dioxide equivalent, compared to 2022. Our RISE strategy also focuses on addressing climate change impacts upstream and downstream in the value chain. This includes improving product energy efficiency and increasing our "handprint"—the ways in which Intel technologies can help others reduce their footprints, including Internet of Things solutions that enable intelligence in machines, buildings, supply chains, and factories, and make electrical grids smarter, safer, and more efficient.

In August 2023, we published our inaugural green bond report, which provides an update on the allocation of the net proceeds of the \$1.3 billion principal amount of senior notes issued in 2022. We are using the proceeds from the green bond offering to fund projects that support our investments in sustainable operations, which can include items such as green buildings, energy efficiency, circular economy and waste management, greenhouse gas emissions reductions, water stewardship, and renewable energy.

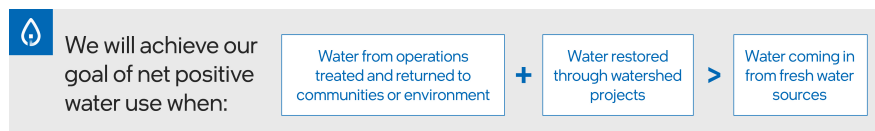
Energy

We focus on reducing our own climate change impact, and over the past two decades have reduced our direct and indirect greenhouse gas emissions associated with energy consumption. Through our RISE goals, we have committed to a goal of conserving 4 billion kWh of energy this decade. We have conserved 1.6 billion kWh¹ of energy cumulatively since 2020. We also invest in renewable electricity and on-site alternative energy projects in support of our 2030 goal to achieve 100% renewable electricity use across our global operations. In 2023, continuing our practice of linking a portion of our executive and employee performance bonus to our corporate sustainability metrics, we linked a portion of the performance bonus to our 2023 target to reach 95% renewable electricity use globally. We reached our target and achieved 99%¹ renewable electricity usage globally in 2023.

¹ This is a preliminary estimate. The final number will be reported in our 2023-24 Corporate Responsibility Report, to be issued later in 2024.

Water Stewardship

Water is essential to the semiconductor manufacturing process. We use ultrapure water to remove impurities from our silicon wafers, and we use fresh and reclaimed water to run our manufacturing facility systems. Through our RISE goals, we have committed to achieve net positive water globally, and as part of that effort, conserve 60 billion gallons of water in this decade. Water conservation reduces the amount of water needed from fresh water sources; we have conserved 35.9 billion gallons¹ of water and enabled restoration of 9.6 billion gallons¹ of water to local watersheds since 2020. In 2023, we linked a portion of our executive and employee performance bonus to our target to conserve and restore 12.0 billion gallons of water during the year.



Circular Economy and Waste Management

We have long been committed to waste management, recycling, and circular economy strategies that enable the recovery and productive re-use of waste streams. Our 2030 goals include a target of zero waste² to landfill, as well as implementation of circular economy strategies for 60% of our manufacturing waste streams in partnership with our suppliers. We continue to focus on opportunities to upcycle waste by improving waste segregation practices and collaborating with our suppliers to evaluate new technologies for waste recovery. In 2023, we linked a portion of our executive and employee performance bonus to our interim target to achieve 5% of waste to landfill.

Governance and Disclosure

We are committed to transparency around our carbon footprint and climate risk, and use the framework developed by the TCFD to inform our disclosure on climate governance, strategy, risk management, and metrics and targets. For governance and strategy, we follow an integrated approach to address climate change, with multiple teams responsible for managing climate-related activities, initiatives, and policies, with senior-level management involvement and board-level oversight, including the Corporate Governance and Nominating Committee. We describe our overall risk management processes in our Proxy Statement, and describe climate-related risks and opportunities in our annual Corporate Responsibility Report, the Intel Climate Change Policy, and "Risk Factors" within this Form 10-K. In addition to what is included in this Form 10-K, information about and progress toward our RISE goals is included in our Corporate Responsibility Report. Our Corporate Responsibility Report also includes a mapping of our disclosure to the TCFD, GRI and SASB frameworks. The Corporate Responsibility Report and our CDP Climate Change Survey are available on our website and are published annually.³ In November 2023, we published our first Climate Transition Action Plan, which is available on our website.

¹ This is a preliminary estimate. The final number will be reported in our 2023-24 Corporate Responsibility Report, to be issued later in 2024.

² Intel defines zero waste as less than 1%.

³ The contents of our website and our Corporate Responsibility Report, Climate Change Policy, Climate Transition Action Plan and CDP Climate Change Survey are referenced for general information only and are not incorporated by reference in this Form 10-K.

Management's Discussion and Analysis

Our Products

We are a global IDM of CPUs and related solutions that we design, develop, manufacture, market, sell, support, and service. Our CPUs and related solutions are incorporated in computing and related end products and services, and utilized globally by consumers, enterprises, governments, and educational organizations. Our customers primarily include OEMs, ODMs, cloud service providers, and other manufacturers and service providers, such as industrial and communication equipment manufacturers and other cloud service providers who buy our products through distributor, reseller, retail, and OEM channels throughout the world. We market and sell these products directly through our global sales and marketing organizations and indirectly through channel partners. We manufacture our products at our fabrication and assembly and test facilities located throughout the world.

Our product offerings provide end-to-end solutions, scaling from data center to network, PCs, edge computing, and the emerging fields of AI and autonomous driving, to serve an increasingly smart and connected world. Products, such as our gaming CPUs, may be sold directly to end consumers, or they may be further integrated by our customers into end products such as notebooks and storage servers. Combining some of these products—for example, integrating FPGAs with Intel Xeon processors in a data center solution—enables incremental synergistic value and performance. In 2023 we launched new products, including the 13th Gen Intel Core mobile processor family, the 4th Gen Intel Xeon Scalable processors with Intel vRAN Boost, and the Intel Core Ultra processors, which feature our first integrated neural processing unit for power-efficient AI acceleration and local inference on the PC.

Our diverse product line includes CPU and chipset, an SoC, or a multichip package based on Intel® architecture that processes data and controls other devices in a system. The primary CPU products in CCG are our Intel Core processors, which include designs specifically for notebook and desktop applications. The primary CPU product in DCAI is our Intel Xeon processor, which includes solutions for data center compute, networking, and the intelligent edge. The primary offerings of NEX include Intel Xeon, Intel Core, and Intel Atom® processor products.

During 2023, we managed our business through the operating segments that are presented below and have included the 2023, 2022 and 2021 financial results for each segment. "Note 3: Operating Segments" within the Notes to Consolidated Financial Statements of this Form 10-K reconciles our segment revenues presented below to our total revenues, and our segment operating income (loss) presented below to our total operating income (loss), for each of the periods presented. We have also included a discussion of our 2023, 2022 and 2021 consolidated results of operations and related information subsequent to the operating segment discussion below.



Overview

We are committed to advancing PC experiences by delivering an annual cadence of leadership products and deepening our relationships with industry partners to co-engineer and deliver leading platform innovation. We engage in an intentional effort to bring together the operating system, system architecture, hardware, and software application integration to enable industry-leading PC experiences. We embrace these opportunities by focusing our roadmap, delivering innovative PC capabilities, and designing advanced PC experiences. By doing this, we believe we help continue to fuel innovation across the industry, providing a solid source of IP, scale, and cash flow for Intel.

Key Business Developments

- We launched our 13th Gen Intel Core mobile and select desktop processors, Intel Core 14th Gen processors, and Intel Core Ultra processors, the first client processor family on Intel 4 technology that features a new neural processing unit to drive AI at scale.
- We launched the industry's first AI PC Acceleration Program to help enable AI on more than 100 million PCs through 2025.
- We worked with industry partners to co-engineer and deliver new experiences with the Intel® Evo™ device, including phone to PC capabilities with Intel® Unison™ application and future premium laptop experiences with AI and Intel Core Ultra processors.

Market and Business Overview

Market Trends and Strategy

In 2023, the PC market started to stabilize from a soft macroeconomic environment and inflationary pressures, with PC supply and demand levels beginning to normalize. We remain positive on the long-term outlook for PCs, as household density is stable to increasing, educational device penetration rates remain low outside of the US, and PC usage remains elevated compared to pre-pandemic rates¹. Commercial growth opportunities also remain as corporations expand the size of their PC fleets, while also replacing older devices. Currently, approximately 200 million commercial devices are more than four years old².

We see the AI PC as a critical inflection point for the PC market over the coming years, and we believe we are well positioned to capitalize on the emerging growth opportunity of AI. We believe that the PC is as essential as ever, and we expect our leadership in client AI to further support our original forecasted long-term PC TAM of approximately 300 million units³. Together with our industry partners, we are working to increase demand and drive market growth through enhanced PC capabilities, which we expect will lead to greater market penetration and faster PC refresh cycles.

As we continue on our strategy to develop more competitive products and more capabilities for customers, we are designing our product roadmap to drive product leadership grounded in a philosophy of openness and choice. We deliver value to our customers by leveraging our engineering capabilities and working with our partners across an open, innovative ecosystem to deliver technology that drives every major vector of the computing experience, including performance, power efficiency, battery life, connectivity, graphics, and form factors, to create the most advanced PC platforms.

Products and Competition

We released our 14th Gen Intel Core desktop processor family, delivering improved single-threaded and multi-threaded performance when gaming, streaming, and recording. In addition, we released our first Intel Core Ultra family of processors, which utilizes a disaggregated architecture and is the first PC platform built on Intel 4 technology. Our latest processor family is the first client processor to feature a dedicated NPU for AI acceleration and delivers improved power efficiency and graphics performance. Intel Core Ultra represents an inflection point in Intel's client processor roadmap as we usher in the age of the AI PC. With new capabilities in the PC, including significantly improved battery life, AI capabilities with our NPU and GPU products, and up to double the graphics performance with Intel Core Ultra processors, we seek to provide compelling reasons to drive refresh in commercial and consumer markets. We expect to deliver more than 230 Intel Core Ultra designs to market.

There is strong demand for our 13th Gen Intel Core processor family, which features Intel® Thread Director technology and our second-generation performance hybrid architecture. We expect to deliver more than 300 designs from partners across major multi-national corporations and leading manufacturers.

We continue to innovate beyond the CPU to deliver premium PC experiences with Intel Evo Edition laptops and Intel vPro Platforms. We worked with industry partners to co-engineer and deliver new experiences with Intel Evo Edition, which are designed to deliver key experience indicators such as responsiveness, battery life, intelligence collaboration, and Intel Unison Multi-Device Experience. Intel vPro is designed for enterprise needs and delivers increased productivity, connectivity, security features, and remote manageability.

We operate in a particularly competitive market. In processors, we compete with Advanced Micro Devices, Inc. (AMD) and vendors who design applications processors based on ARM architecture*, such as Qualcomm Inc. (Qualcomm), and Apple Inc. (Apple), with its M1 and M2 products. We expect this competitive environment to continue to intensify in 2024.

We remain committed to creating an open ecosystem to foster growth and technology innovations. We embrace and collaborate with a global ecosystem of industry partners to deliver leadership technologies together. We launched the industry's first AI PC Acceleration Program, designed to provide the software ecosystem with engineering tools and resources to enable AI on more than 100 million PCs through 2025. We also announced a collaboration with Microsoft to drive the development of AI on personal computing, with Intel Core Ultra processors and Windows 11 expected to scale across the ecosystem of Intel and Microsoft OEM and ISV partners.

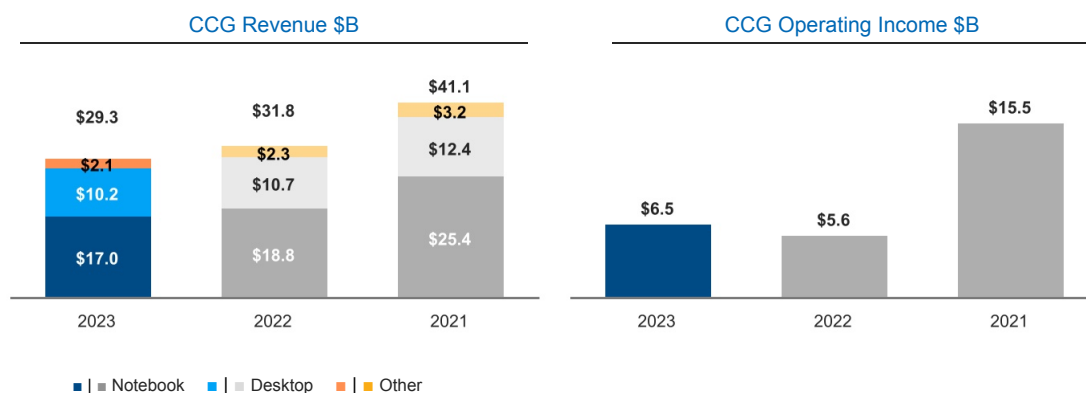
During 2023, we continued to diversify our product strategy across nodes, advance our packaging capabilities with disaggregated silicon, and balance internal and external manufacturing. We increased our levels of operating inventory and worked with our customers to develop strategically located supply hubs that forward position inventory. In addition, we continue to invest in a globally diverse supply chain that enables us the flexibility and proximity to support customers. These have further enhanced the service and responsiveness we are able to provide our customers.

¹Source: Intel calculated PC density from industry analyst reports.

²Source: Intel calculated volume of devices over four years old from industry analyst reports and internal data.

³Source: Intel calculated multi-year TAM forecast derived from industry analyst reports.

Financial Performance



Revenue Summary

2023 vs. 2022

- Notebook revenue was \$17.0 billion, down \$1.8 billion from 2022. Notebook volume decreased 5% from 2022, driven by lower demand across market segments, partially offset by increased volume in the second half of the year as customer inventory levels normalized compared to higher levels in the first half. Notebook ASPs decreased 5% from 2022 due to relative strength in the education market segment resulting in a higher mix of small core products combined with a higher mix of older generation products.
- Desktop revenue was \$10.2 billion, down \$495 million from 2022. Desktop volume decreased 9% from 2022, driven by lower demand across market segments, partially offset by increased volume in the second half of the year as customer inventory levels normalized compared to higher levels in the first half. Desktop ASPs increased 5% from 2022, due to an increased mix of product sales to the commercial and gaming market segments.
- Other revenue was \$2.1 billion, down \$229 million from 2022, primarily driven by the continued ramp down of our legacy smartphone modem business and lower demand for our wireless and connectivity products as a result of lower notebook volumes.

2022 vs. 2021

- Notebook revenue was \$18.8 billion, down \$6.7 billion from 2021. Notebook volume decreased 36% from 2021, driven by lower demand in the consumer and education market segments, and notebook ASPs increased 15% from 2021 due to an increased mix of commercial and consumer products and a lower mix of education products.
- Desktop revenue was \$10.7 billion, down \$1.8 billion from 2021. Desktop volume decreased 19% from 2021, driven by lower demand in the consumer and education market segments, and desktop ASPs increased 5% from 2021, primarily from an increased mix of commercial products.
- Other revenue was \$2.3 billion, down \$870 million from 2021, primarily driven by the continued ramp down of our legacy smartphone modem business and lower demand for our wireless and connectivity products.

Operating Income Summary

Operating income increased 17% year over year, and operating margin was 22% in 2023 and 18% in 2022.

(In Millions)

\$	6,520	2023 Operating Income
1,692		Lower period charges driven by the sell-through of previously reserved inventory and lower reserves taken in 2023
1,220		Lower operating expenses driven by various cost-cutting measures
268		Lower period charges primarily driven by a decrease in product ramp costs
(1,704)		Lower product margin primarily from lower notebook and desktop revenue
(385)		Higher unit cost primarily from increased mix of Intel 7 products
(140)		Higher period charges related to excess capacity charges
\$	5,569	2022 Operating Income
(3,047)		Lower product margin from notebook revenue
(2,183)		Higher notebook and desktop unit cost primarily from increased mix of Intel 7 products
(1,306)		Lower product margin from desktop revenue
(1,400)		Higher operating expenses driven by increased investments in leadership products
(1,155)		Higher period charges primarily driven by inventory reserves taken in 2022
(364)		Lower CCG other product margin driven by lower demand for our wireless and connectivity products and the continued ramp down from the exit of our 5G smartphone modem business
(267)		Higher period charges primarily associated with the ramp of Intel 4
(162)		Higher period charges related excess capacity charges
192		Lower period charges due to a benefit related to insurance proceeds received for business interruption and property damage that occurred in 2020
(262)		Other
\$	15,523	2021 Operating Income



Data Center and AI

Overview

DCAI delivers cutting-edge workload-optimized solutions to cloud service providers and enterprises, along with silicon devices for communications service providers, network and edge, and HPC customers. Our unique capabilities enable us to help solve our customers' most complex challenges with the depth and breadth of our hardware and software portfolio, advanced packaging, and at-scale manufacturing made possible through a resilient, global supply chain. Our global customers and partners encompass cloud hyperscalers, multinational corporations, small-and medium-sized enterprises, independent software vendors, systems integrators, communications service providers, and governments.

Key Business Developments

- We have sold more than 2 million 4th Generation Intel Xeon Scalable processors as of the end of 2023, and launched the 5th Gen Intel Xeon processors in Q4 2023.
- We were recognized by MLCommons, which published MLPerf Training performance benchmark data showing that 4th Gen Xeon and Intel Gaudi 2 are two compelling, open computing platforms in the AI market that compete on performance, price, and broad availability.
- We announced our intent to separate PSG into a standalone company, giving PSG the autonomy and operational and financial flexibility it needs to accelerate growth to more effectively compete in the FPGA market. Standalone financial reporting for PSG began January 1, 2024.

Market and Business Overview

Market Trends and Strategy

Data is a significant force in society and is generated daily at an unprecedented pace. The desire to harness insights from data to drive better outcomes for businesses and society is ever expanding. AI is becoming pervasive in nearly all applications, creating the potential for intelligence everywhere, and enabling powerful new uses of compute resources across all market segments. The installed base of Intel Xeon processors, combined with our portfolio of heterogeneous compute solutions (FPGAs, GPUs, IPUs, and AI accelerators), we believe positions us to lead in this high-growth area. DCAI is integral to our growth in AI through deep investments in the AI ecosystem, developer tools, frameworks, networking and memory, technologies, and open standards to drive a scalable path forward.

We take a system-level approach that supplies the necessary hardware and software optimized for power and performance. Our technology is differentiated at the system level and in high-growth workloads based on our integrated hardware acceleration engines and software. For example, architected into our Intel Xeon processors are Intel® Advanced Matrix Extensions (Intel® AMX) for AI acceleration; Intel® Software Guard Extensions (Intel® SGX), providing enclaves of protected memory designed to deliver enhanced security for sensitive data; and Intel® Crypto Acceleration, which is designed to deliver breakthrough performance across cryptographic algorithms. We believe this acceleration and performance will continue to drive our differentiated value and growth across our customer base.

Products and Competition

Our products and services include:

- A portfolio of hardware, including Intel Xeon processors, the Intel® Xeon CPU Max Series, the Intel® Data Center GPU Max Series, the Intel® Data Center GPU Flex Series, Intel Agilex® and Intel® Stratix® FPGAs, Intel® eASIC™ devices, and the Intel Gaudi processors.
- Platform enabling and validation in partnership with ODMs, OEMs, CSPs, and independent software vendors.
- Optimized solutions for leading workloads such as AI, cryptography, security, storage, and networking, leveraging differentiated features supporting diverse compute environments.

We provide our customers with an extensive portfolio of silicon and software products, engineered to deliver workload-optimized performance. Our hardware portfolio comprises CPUs, GPUs, domain-specific accelerators, and FPGAs, designed to support the performance, agility, and security that our customers demand. Deployment of our silicon platforms is accelerated by a software development environment that enables workload mobility across our heterogeneous architectures and enables developers to execute their workloads on the hardware that best meets application requirements.

Our competitors include AMD, providers of GPU products such as NVIDIA, companies developing their own custom silicon, and new entrants and incumbents developing ARM- and RISC-V-based products customized for specific data center workloads. We expect this competitive landscape to continue.

The Intel Xeon Scalable processor family delivers advanced CPUs for the data center, the network, and the edge, driving industry-leading performance, manageability, and security with differentiated features and capabilities. In 2023, we launched the 5th Gen Intel Xeon Scalable processors, which utilize modular SoCs for increased scalability and flexibility to deliver a range of products that meet the growing scale, processing, and power efficiency needs for AI, cloud, and enterprise installations.

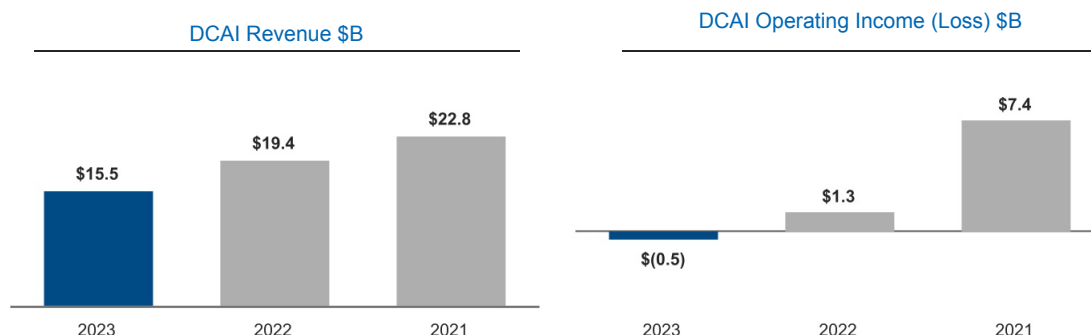
Our Intel Gaudi 2 AI deep learning processor is engineered to allow customers to benefit from the most cost-effective, high-performance training and inference alternative to comparable GPUs.

Our FPGA and structured ASIC portfolio enhances Intel's ability to meet the needs of customers in the data center, across the network, and at the edge. In 2023, we released 21 new products within our FPGA business. These include the Intel Agilex 9 and Intel Agilex 7 product families, which are intended to handle the increased demand for customized workloads, including enhanced AI capabilities, and to provide lower total cost of ownership.

Our Intel Xeon CPU Max Series, introduced in 2022, was the first and only x86-based processor with high bandwidth memory for HPC and AI workloads. The Intel Data Center GPU Max Series 1550 and 1100 are now broadly available through OEM systems. Our Intel CPU Max Series includes 64GB of on-package high bandwidth memory, which benefits data-intensive HPC workloads. Our Intel Data Center GPU Flex Series offers customers a flexible, general-purpose GPU for the data center and the intelligent visual cloud. We delivered and installed, in collaboration with Argonne National Laboratory (ANL) and Hewlett Packard Enterprise (HPE), the largest cluster of Max GPUs to the Aurora Supercomputer at ANL.

The ubiquity of Intel Xeon in the installed base, along with our heterogeneous compute solutions combined with software that unlocks the value of our hardware, enable our customers to develop highly differentiated solutions. Our integrated approach has created significant value for Intel, our customers, and our partners by helping us mitigate risks, reduce costs, build brand value, and identify new market opportunities to apply our technology to address our customers' and society's most complex issues.

Financial Performance



Revenue Summary

2023 vs. 2022

Revenue was \$15.5 billion, down \$3.9 billion from 2022, driven by a decrease in server revenue. Server volume decreased 37% from 2022, due to lower demand in a softening CPU data center market. Server ASPs increased 20% from 2022, primarily due to a lower mix of hyperscale customer-related revenue and a higher mix of high core count products.

2022 vs. 2021

Revenue was \$19.4 billion, down \$3.3 billion from 2021, due to a decrease in server revenue, partially offset by higher other DCAI revenue. Server volume decreased 16% from 2021, led by enterprise customers in a competitive environment, and due to customers tempering purchases to reduce existing inventories in a softening data center market. Server ASPs decreased 5% from 2021, driven by a higher mix of revenue from hyperscale customers. Other DCAI revenue increased 18% from 2021 primarily driven by growth in our FPGA business.

Operating Income (Loss) Summary

We had an operating loss of \$530 million in 2023, compared to operating income of \$1.3 billion in 2022.

(In Millions)	
\$ (530)	2023 Operating Income (Loss)
(2,709)	Lower product margin primarily due to lower server revenue
(1,269)	Higher server unit cost primarily from increased mix of Intel 7 products
(171)	Higher period charges related to excess capacity charges
1,337	Lower operating expenses driven by various cost-cutting measures
520	Lower period charges primarily driven by a decrease in product ramp costs
462	Lower period charges driven by the sell-through of previously reserved inventory and lower reserves taken in 2023
\$ 1,300	2022 Operating Income
(3,325)	Lower product margin from server revenue
(1,137)	Higher period charges primarily associated with the ramp up of Intel 4
(1,043)	Higher operating expenses driven by increased investments in leadership products
(671)	Higher server unit cost from increased mix of 10nm SuperFin products
(645)	Higher period charges driven by inventory reserves taken in 2022
(189)	Higher period charges related to excess capacity charges
785	Higher product margin from DCAI other product revenue
223	Lower period charges due to a benefit related to insurance proceeds received for business interruption and property damage that occurred in 2020
(74)	Other
\$ 7,376	2021 Operating Income



Overview

NEX transforms the world's networks and edge compute systems from fixed-function hardware to general-purpose compute, acceleration, and networking devices running cloud native software on programmable hardware. We work with partners and customers to deliver and deploy intelligent edge platforms that allow developers to achieve agility and to drive automation using AI for efficient operations while securing the integrity of their data at the edge. We have a broad portfolio of hardware and software platforms, tools, and ecosystem partnerships for the rapid digital transformation happening from the cloud to the edge. We are leveraging our core strengths in compute, connectivity, software, and manufacturing at scale to grow traditional markets and to accelerate entry into emerging ones.

Key Business Developments

- We launched the 4th Gen Intel Xeon processor with Intel vRAN Boost, and announced the Intel Xeon D-1800 series and the Intel Xeon D-2800 series processors optimized for cloud, edge, and 5G networks.
- We continue to update solutions to improve developers' digital strategies and to accelerate market adoption of edge and AI applications. We announced 13th Gen Intel Core processors for Internet of Things edge and OpenVINO toolkit version 2023.1 with enhanced features for GenAI and edge computing.
- We continue to work with our ecosystem partners like Ericsson, Nokia, Cisco, Dell Technologies, HPE, Lenovo, Amazon, Google, and Microsoft to drive the software-defined transformation of the world's network and edge infrastructure and accelerate AI-driven automation of physical operations.